

MODEL ANSWER SHEET — MATHEMATICS MID-TERM 2025

BARCODE
NS-MATH-2025

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Q1(i) — Maximum Marks: 3

Award Marks Criteria:

- A computer system is an electronic machine that accepts data (input), processes it according to instructions, stores it, and produces meaningful output.
- Four main components: (1) Input Unit — accepts data from user. (2) Processing Unit (CPU) — executes instructions. (3) Memory Unit — stores data and programs. (4) Output Unit — displays results.
- Award: 1 mark for definition + 0.5 mark each for any 4 components correctly named and explained = 3 marks.

Q1(ii) — Maximum Marks: 3

Award Marks Criteria:

- RAM (Primary): volatile, fast, expensive, directly accessed by CPU. Examples: DDR4 RAM, SRAM.
- Hard Disk (Secondary): non-volatile, slower, cheaper, stores data permanently. Examples: HDD, SSD.
- Award: 1 mark per correct point with example. Any valid examples accepted.

Q1(iii) — Maximum Marks: 3

Award Marks Criteria:

- OS Definition: Software that manages computer hardware and software resources and provides services for programs.
- Functions (any 3): Process management, Memory management, File system management, Device management, Security and access control, UI (GUI/CLI).
- Award: 0.5 mark for definition + 0.5 mark for each function (up to 5 functions) with brief explanation.

Q1(iv) — Maximum Marks: 3

Award Marks Criteria:

- Multiprogramming: Multiple programs loaded in memory simultaneously. CPU switches between them when one waits for I/O. Increases CPU utilisation.
- Multitasking: Multiple tasks appear to run simultaneously by rapidly switching CPU between them. Example: Playing music while browsing.
- Award: 1 mark each for correct definition + 0.5 mark each for example. Full marks for any valid explanation.

Q2(i) — Maximum Marks: 4

Award Marks Criteria:

- Diagram must show: CPU (containing CU + ALU + Registers), Main Memory, Input devices, Output devices, connected by buses (Data bus, Address bus, Control bus).
- Labels required: Control Unit, ALU, Memory, Input, Output, System Bus.
- Award: 2 marks for correct labelled diagram + 1 mark per unit explanation (CU + ALU + Memory = 2 marks).

Q2(ii) — Maximum Marks: 4

Award Marks Criteria:

- Fetch: PC sends address to MAR → Memory fetches instruction → placed in MDR → copied to IR → PC incremented.
- Decode: CU decodes instruction in IR, identifies opcode and operands.

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- Execute: ALU performs operation, result stored in register or memory.
- Award: 1 mark per stage (Fetch/Decode/Execute) with correct detail + 1 mark for correct flowchart representation.

Q3(i) — Maximum Marks: 3**Award Marks Criteria:**

- 255 in Binary: 11111111 (Award 1 mark)
- 255 in Octal: 377 (Award 1 mark)
- 255 in Hexadecimal: FF (Award 1 mark)
- Show conversion steps for full credit. Final answer without steps: 0.5 mark each.

Q4(i) — Maximum Marks: 5**Award Marks Criteria:**

- Classification by Size: Supercomputer (weather forecasting), Mainframe (banking), Minicomputer (mid-range server), Microcomputer (PC, laptop), Embedded (ATM, washing machine).
- Classification by Purpose: General purpose (home PC), Special purpose (ATM, traffic controller).
- Award: 0.5 mark per category correctly named + 0.5 mark for valid example. Maximum 5 marks.

Q7(i) — Maximum Marks: 5**Award Marks Criteria:**

- SRAM: Uses flip-flops, faster, more expensive, no refresh needed, used in cache memory. Power consumption higher per bit.
- DRAM: Uses capacitors, slower, cheaper, needs periodic refresh, used in main memory (RAM sticks).
- Comparison table (if drawn): 2 marks for correct comparison + 1.5 marks per type = 5 marks.
- Accept any valid comparison. Missing refresh explanation: deduct 0.5 mark.